

Performance Testing Log

Date:	15 March 2026	Team ID:	PNT2022TMID124356
Project Name:	ANTI DEBT	Lead:	Lakshmi Sravya Kagitha (Team Lead)

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman / Custom performance measuring timing scripts
Type of Testing	Load Testing, Response Latency Profiling
Target Module / API	Authentication login, Loans list retrieve, DTI ratios calculations, Gemini Recommendations API
Test Environment	Local Development Server / SQLite file database
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Concurrent login authentications (JWT token generation check)	20	10s	All tokens generated; response time < 500ms.
2	Concurrent loan ledger CRUD entries read/write	50	20s	No database locks; transactions complete successfully.
3	FastAPI DTI dashboard metrics recalculations audit	50	15s	All math checks return under 100ms.
4	Concurrent Gemini settlement recommendations JSON generation	5	30s	API resolves correctly; response time < 3 seconds.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	180 ms (Local APIs) / 1.6s (Gemini)	Pass	Highly optimized FastAPI routes.
2	Response Time (Max)	< 5 seconds	350 ms (Local APIs) / 2.8s (Gemini)	Pass	Complies with limits.
3	Throughput (Req/sec)	200 req/sec	240 req/sec	Pass	Lightweight async backend.

4	Error Rate	< 1%	0.0%	Pass	Zero transaction failures during concurrency tests.
5	CPU Utilization	< 80%	12.4%	Pass	Minimum overhead on development machine.
6	Memory Utilization	< 80%	45 MB (Python App) / 95 MB (Node)	Pass	Extremely lightweight profile footprint.